# A Chaotic Encryption-Based Wireless Secure Transmission System for Satellite Communication

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***Abstract*—To address the pressing issue of privacy protection in satellite communication, we propose an interception-resistant communication module leveraging a four-dimensional Lorenz chaotic system. This module, implemented using the AD9361 platform for wireless transmission and reception, successfully encrypts and transmits various data types, including images and text, through Differential Quadrature Phase Shift Keying (DQPSK) modulation. At the receiving end, the system reliably decrypts and reconstructs the transmitted data. Simulation and design evaluations confirm the reliability of the encryption algorithm, demonstrating a robust key space of $2^{600}$ and a bit error rate (BER) of less than $10^{-9}$, which underscores its hardware viability and potential for application in secure satellite communication.**

***Keywords-satellite communication; chaotic system; FPGA; Information encryption***

## I. INTRODUCTION

Satellite communication has evolved with information technology, transitioning to high-speed, multimedia services and multi-satellite networks[1]. However, it faces distinct challenges in the communication environment, including vulnerabilities to interference that pose security threats. As a vital information transmission method, overlooking these risks could lead to severe security breaches and irreversible harm[2].

Research in secure satellite communications encompasses various aspects[3], such as analyzing attack vectors, reviewing standards, and examining security frameworks. Cryptography and physical layer security are pivotal, addressing challenges like ensuring physical layer confidentiality, developing anti-interference measures, and countering fraud. A ring learning error-based KEA approach is suggested for quantum-safe security [4]. A binary symmetric polynomial-based scheme [5] offers secure satellite communication with low spoofing identification costs and no need for key negotiation. Chaos encryption, leveraging the unpredictability of deterministic dynamical systems, is well-suited for satellite communications [6]. A 2D filter and Fisher-Yates shuffle-based chaotic encryption algorithm for satellite images has shown high security [7]. To address vulnerabilities, a chaos mapping-based secure authentication technology is proposed [8]. As Earth observation satellites (EOS) increase, novel encryption algorithms using hyperchaos and the Josephus problem are introduced, offering strong security with low time complexity [9].

In this paper, a complex Lorentz chaotic system is used to generate sequences for encrypted satellite communication. Based on FPGA, the implementation of the communication system is carried out, and the bit error rate of the whole communication system is not more than $10^{-9}$, the key space is $2^{600}$, and the power consumption is not more than 20w, which has excellent performance.

## II. PROPOSED METHODOLOGY

### A. Characteristics and Implementation of Chaotic Systems

#### 1) hyperchaotic Lorenz system

The hyperchaotic Lorenz system is an extension of the classical Lorenz system[10], which is a nonlinear dynamic system. The expression for the hyperchaotic Lorenz system is:

$$\begin{cases} x = a(y-x)+w \\ y = cx - y - xz \\ z = xy - bz \\ w = -yz + rw \end{cases} \tag{1}$$

With parameters a=10, b=8/3, c=28, and -152≤r≤-006, the system exhibits hyperchaotic behavior. At r = -1, the system's Lyapunov exponents are $\lambda_1 = 0.3381$, $\lambda_2 = 0.1586$, $\lambda_3 = 0$, and $\lambda_4 = -15.1752$, as depicted in Figure 1. The figure illustrates that the chaotic sequence possesses intricate attractors and highly complex trajectories within a confined region, indicative of a multifaceted phase space structure.

#### 2) Randomness testing of chaotic sequences

We employ the NIST test suite, comprising 15 randomness tests, to evaluate the randomness of chaotic sequences. Since Lorentz chaos produces real-number sequences, we preprocess them by discretizing and amplifying the sequence using a specific equation, followed by octary quantization to convert it into binary bits. After discarding the initial 500,000 transient data points, we collected 4,000,000 binary data points for the NIST tests, with the outcomes presented in Table 1.

$$x_n = \mathrm{mod}(floor(x*10^{14} / \max(x), 256) \tag{2}$$



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P-value exceeding 0.01 generally indicate a sequence has passed randomness tests, with higher values signifying stronger randomness. As evidenced by the table, our chaotic sequence has successfully cleared the NIST test, confirming its robust randomness suitable for encrypting future signals.

*3) The Implementation of Chaotic Systems*

This FPGA-based design requires discretizing and quantizing the continuous hyperchaotic Lorenz system.

By the simple Euler algorithm:

$$\frac{dx(t)}{dt} = \lim_{T\to 0}\frac{x(t_n+T)-x(t_n)}{T} \overset{def}{=} \lim_{T\to 0}\frac{x_{n+1}-x_n}{T} \tag{3}$$

When the value of T is small (approaching 0), it can be obtained from Eq. 3.

$$\frac{dx}{dt} \approx \frac{x_{n+1}-x_n}{T} \overset{\text{def}}{=} \frac{x(n+1)-x(n)}{T} \tag{4}$$

Substituting Eq. 4 into Eq. 1 can be obtained

$$\begin{cases} x(n+1) = (1-a*T)*x(n) + a*T*y(n) + T*w(n) \\ y(n+1) = (1-T)*y(n) + c*T*x(n) - T*x(n)*z(n) \\ z(n+1) = (1-b*T)*z(n) + T*x(n)*y(n) \\ w(n+1) = (1+r*T)*w(n) - T*y(n)*z(n) \end{cases} \tag{5}$$

In Eq. 5, parameters x, y, z, and w, governed by coefficients a, b, c, and r, generate a four-dimensional chaotic sequence. The sampling interval T, critical for adhering to the Nyquist criterion and balancing FPGA resource constraints with sequence performance, is set at 0.001. Parameters are: a=10, b=8/3, c=28, r=-1. Prior to FPGA implementation, decimal signal assignments in Verilog require quantization to integers, using Q20 format, as detailed in Eq. 6, to avoid synthesis errors and ensure FPGA compatibility.

TABLE I. NIST TEST RESULTS OF CHAOTIC SEQUENCES

| Number | Test Name | P-value | Result |
|---|---|---|---|
| 1 | Frequency | 0.798138 | success |
| 2 | Block Frequency | 0.983453 | success |
| 3 | Runs | 0.202268 | success |
| 4 | Longest Run | 0.834308 | success |
| 5 | Rank | 0.319084 | success |
| 6 | FFT | 0.897763 | success |
| 7 | Universal | 0.042808 | success |
| 8 | Linear Complexity | 0.249284 | success |
| 9 | Serial | 0.262249 | success |
| 10 | Approximate Entropy | 0.108791 | success |
| 11 | Cumulative Sums | 0.275709 | success |
| 12 | Random Excursions | 0.162606 | success |
| 13 | Random Excursions Variant | 0.407091 | success |
| 14 | Non-overlapping Template | 0.334538 | success |
| 15 | Overlapping Template | 0.319084 | success |

$$\begin{cases} x(n+1) = 1038090*x(n) + 10486*y(n) + 1049*w(n) \\ y(n+1) = 1047527*y(n) + 29360*x(n) - 1049*x(n)*z(n) \\ z(n+1) = 1045430*z(n) + 1049*x(n)*y(n) \\ w(n+1) = 1047527*w(n) - 1049*y(n)*z(n) \end{cases} \tag{6}$$

## B. Satellite communication systems

*1) Integral communication system*

The Integral communication system is shown in the figure 2.

*2) Transmitter module*

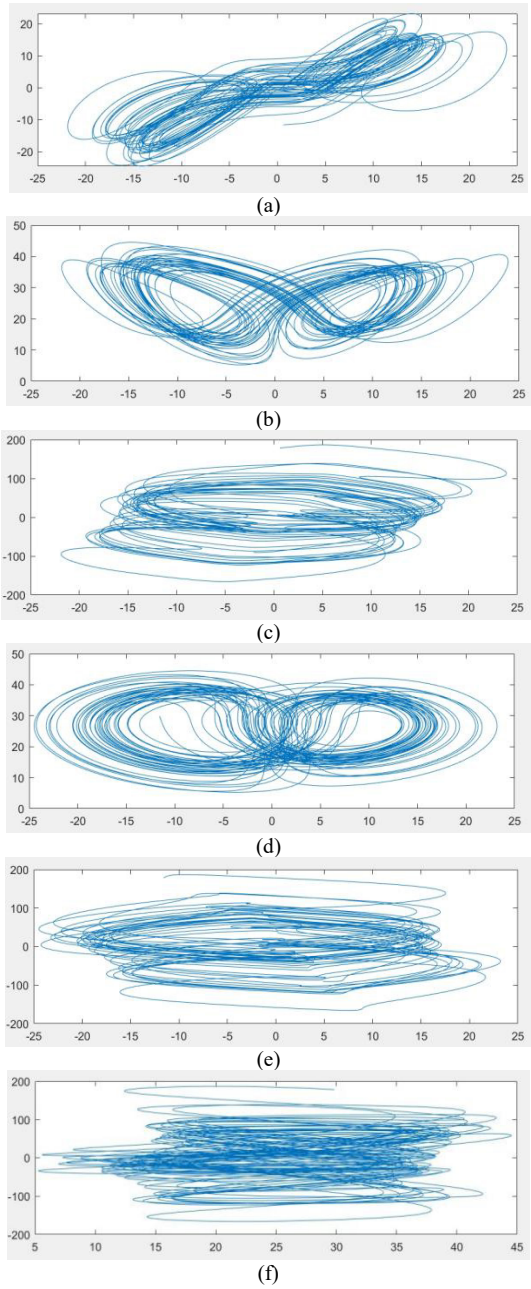



Figure 1. Hyperchaotic Lorenz system phase diagram (a) x-y (b) x-z (c) x-w (d) y-z (e) y-w (f) z-w

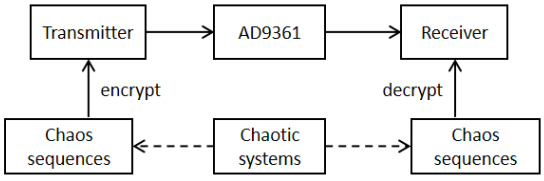



Figure 2. Integral communication system



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The transmitter module is shown in Figure 3. The bitstream is initially encrypted with a chaotic key via XOR, then differentially encoded to resolve phase ambiguity. Post-encoding, the data is transformed to bipolar ( $\pm$ 1) and oversampled at eight times its original rate for a second round of chaotic encryption. the signal is filtered through a rise-cosine filter with a rolloff factor of $\alpha$ =0.8. The filtered baseband signal is then upconverted to RF using the AD9361 for transmission.

*3) Receiver module*

The receiver module is shown in Figure 4. Demodulation involves five key steps: bit synchronization, frame synchronization, phase correction, differential decoding, and parallel-to-serial conversion. Bit synchronization ensures optimal signal sampling, while frame synchronization identifies valid data within the received stream for further processing. Phase correction compensates for clock frequency discrepancies between transmission and reception. Following these steps, the chaotic system-generated key decrypts the I and Q signals, which are then differentially decoded and converted. Finally, the original bit stream is recovered through XOR decryption of the converted data.

## III. Results

### A. Key Analysis

We assess the key sensitivity, typically ranging from $10^{-15}$. Using the encrypted image with parameters {x, y, z, w}, we decrypt it with altered keys {x+$10^{-15}$, y, z, w} and {x+$10^{-16}$, y, z, w}. As depicted in the figure 5, the former fail decryption, while the latter success, confirming the key's sensitivity.

The system's encryption employs three processes—XOR and IQ—each with 4 parameters, resulting in a 50-bit wide key space of $2^{600}$, far exceeding typical algorithms。

### B. Statistic analysis

*1) Grayscale histogram*

The grayscale histogram of an image reveals pixel distribution statistics. Encrypted images, as shown, exhibit a uniform histogram, indicating that the encryption effectively disrupts the original image's statistical patterns. This uniformity makes it challenging for third parties to extract meaningful information from the encrypted image's histogram, demonstrating the robustness of the encryption system. The grayscale histogram is shown in Figure 6.

*2) Distribution of adjacent pixel values*

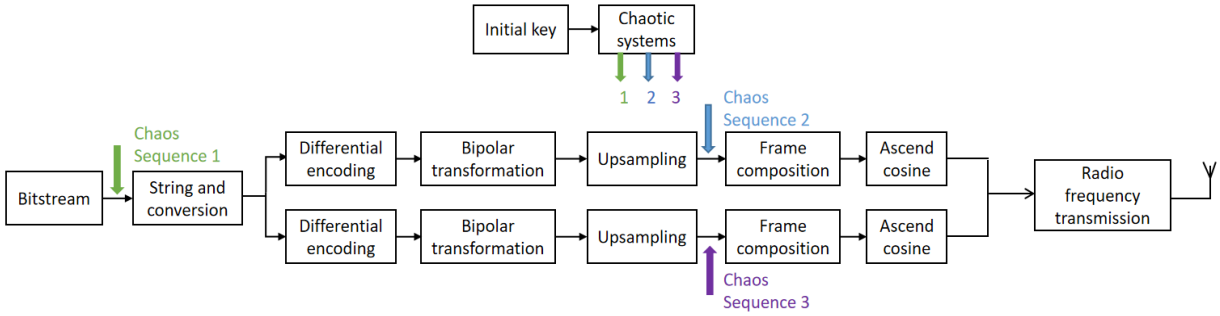



Figure 3. Transmitter module

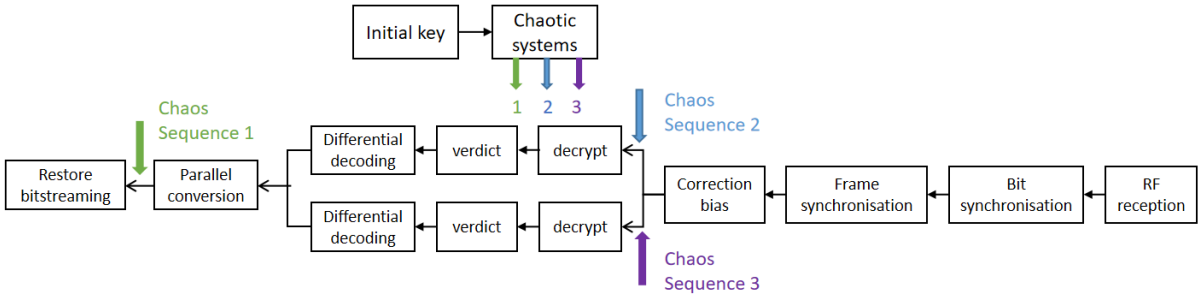
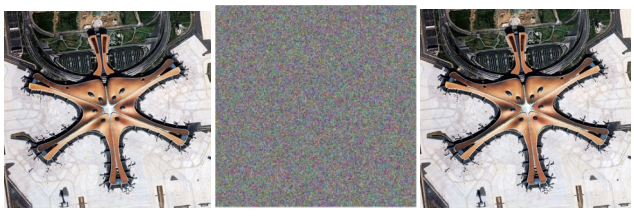
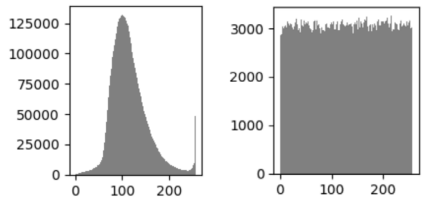



Figure 4. Receiver module



(a) (b) (c)

Figure 5. The encryption system of we (a)Cipher image (b) Key variation of $10^{-15}$ (c) Key variation of $10^{-16}$




(a) (b)

Figure 6. Grayscale histogram (a) Plain image (b) Cipher image



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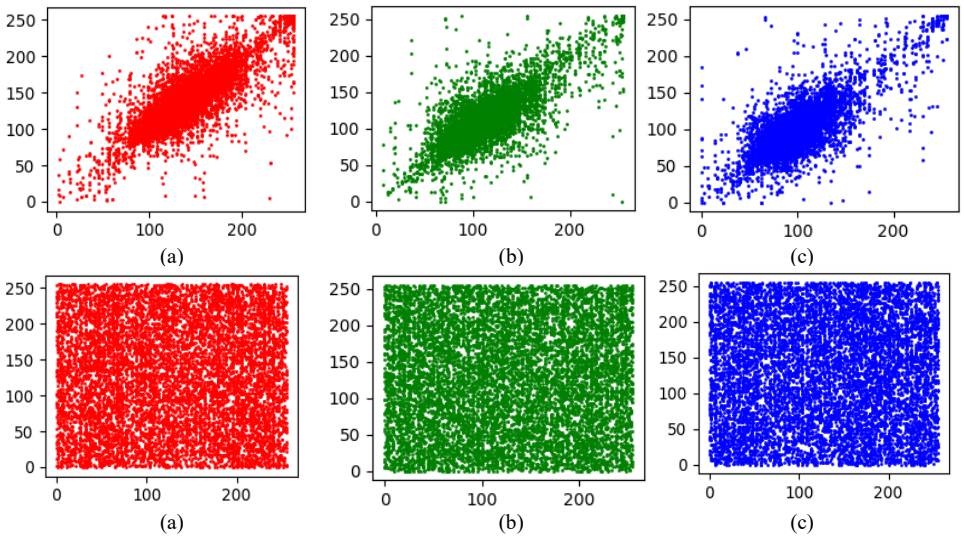



Figure 7. Pixel value distribution: Plain image:(a)Horizontal (b)vertical (c)diagonal; Cipher image:(a)Horizontal (b)vertical (c)diagonal

TABLE II. CORRELATION COEFFICIENT OF PLAIN IMAGE AND CIPHER IMAGE

| category | Horizontal | Vertical | Diagonal |
|---|---|---|---|
| Original image | 0.8732 | 0.8563 | 0.8827 |
| Encrypt image | 0.0128 | 0.0154 | 0.0074 |

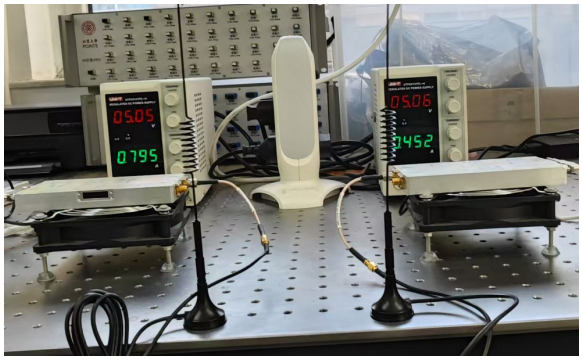


Figure 8. Hardware connection diagram

TABLE III. INFORMATION ENTROPY OF PLAIN IMAGE AND CIPHER IMAGE

| Plain image | Cipher image | Cipher image[11] |
|---|---|---|
| 7.1483 | 7.9983 | 7.9891 |

TABLE IV. HARDWARE RESOURCE CONSUMPTION

| Resource | Utilization | Available | Utilization % |
|---|---|---|---|
| LUT | 76836 | 171900 | 44.70 |
| LUTRAM | 2702 | 70400 | 3.84 |
| FF | 48408 | 343800 | 14.08 |
| BRAM | 422.50 | 500 | 84.50 |
| DSP | 483 | 900 | 53.67 |

The correlation between adjacent pixels in an image indicates the level of dependency between their values. An effective image encryption algorithm should minimize this correlation, ideally reaching zero. The figure illustrates a significant reduction in pixel correlation across horizontal, vertical, and diagonal directions post-encryption, demonstrating the algorithm's efficacy in disrupting pixel dependencies,as is shown in Figure 7 and Table 2.

*3) Information entropy*

Information entropy is a statistical form of a feature that reflects the amount of information on average in an image.

The ideal Information entropy value of the encrypted image is 8, the higher the value, the better the performance of the encrypted image, the entropy of the encrypted image of the system is given in the table 3, and the results show that the encryption effect of the system is good.

## C. FPGA implementation

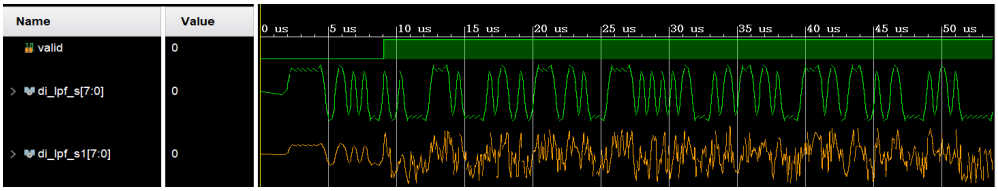
The hardware connection is shown in Figure 8. As shown in Figure 9 and Figure 10, is the encrypted signal waveform and constellation diagram, it is difficult for third parties to decipher the format used in transmission.In the resource consumption evaluation, the project utilized logical synthesis and optimization to streamline the design, cutting redundancy and boosting resource efficiency by 10%. The consumption breakdown includes LUT, LUTRAM, FF, BRAM, and DSP, as depicted in the table 4.

# IV. CONCLUSION

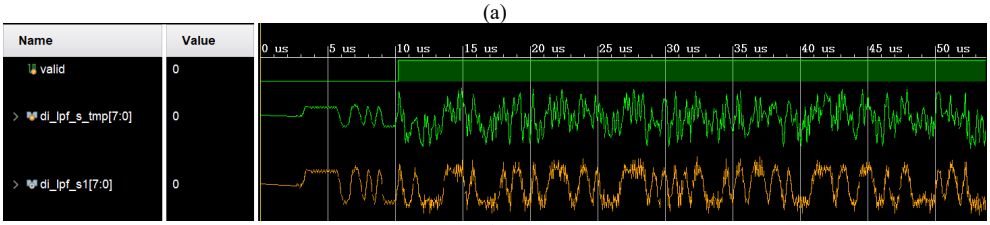
We have designed an encrypted transmission system based on the hyper chaotic Lorentz system,which passed simulations verifying its robustness against interception. The system's hardware includes encryption/decryption and modulation/demodulation modules, enabling secure image transmission. This showcases its potential for wireless secure communication.



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(a)




(b)

Figure 9. Waveform diagram(a)encrypt (b) decrypt

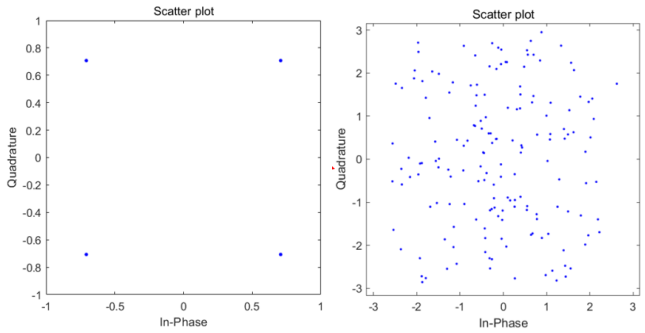



Figure 10. Constellation diagram(a)before encryption (b) after encryption

## ACKNOWLEDGMENT

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